RITUXIMAB PATIENT KNOWLEDGE ASSISTANT

AI-Powered RAG System for Rituximab Therapy Information

Project Report · Raju Kumar · 2025

Field	Details
Project Title	Rituximab Patient Knowledge Assistant
Version	Phase 1 — Production Release
Type	AI / NLP — Retrieval-Augmented Generation (RAG)
Domain	Healthcare — Oncology, Rheumatology, Neurology
Author	Raju Kumar
Year	2025
Live URL	https://rituximab-rag-assistant.onrender.com
Vector Database	Pinecone Serverless (AWS us-east-1)
Embedding Model	OpenAI text-embedding-3-small (1536 dimensions)
LLM	GPT-4o-mini via OpenAI API
Web Framework	FastAPI + Uvicorn
Deployment	Render.com — GitHub CI/CD auto-deploy
GitHub Repository	github.com/raju-AI-portfolio/rituximab-rag-assistant
Vectors in Pinecone	1,019 dense vectors
Knowledge Sources	FDA, NIH/PubMed, NCCN, MedlinePlus, ClinicalTrials.gov

1. Executive Summary

The Rituximab Patient Knowledge Assistant is an AI-powered information system designed to help patients undergoing Rituximab (Rituxan) therapy access accurate, structured, and source-verified medical information at any time of day. Built using Retrieval-Augmented Generation (RAG) architecture, the system retrieves relevant content from a curated knowledge base of official medical documents — including FDA prescribing labels, NIH MedlinePlus patient education content, NCCN clinical guidelines, and PubMed research abstracts — before generating grounded, patient-friendly responses.

The project was developed in 6 structured steps: document collection, text processing and chunking, vector database construction, RAG pipeline development, terminal chat interface, and a browser-based web UI. The completed application was successfully deployed as a live public web service on Render.com.

2. Problem Statement

Rituximab is one of the most widely prescribed biologic therapies globally, used across multiple serious conditions including Non-Hodgkin Lymphoma (NHL), Chronic Lymphocytic Leukemia (CLL), Rheumatoid Arthritis (RA), Multiple Sclerosis, and Lupus. Despite its widespread use, patients consistently face significant challenges accessing timely, accurate information about their therapy.

2.1 Core Challenges

- Complex therapy: Rituximab involves lengthy infusions, strict premedication protocols, serious safety warnings, and varies by indication
- Patient anxiety: Patients experience fear and confusion about side effects, infusion reactions, and long-term immune suppression
- Information gap: Medical teams are unavailable 24/7 to answer routine questions, leaving patients to rely on unreliable online sources
- Non-adherence: Poor patient education is a primary driver of treatment non-adherence, leading to worse outcomes and higher healthcare costs
- Misinformation risk: Generic AI chatbots hallucinate medical information, making them unsafe for clinical use without grounding in authoritative sources

3. Project Objectives

- Design a domain-specific RAG system grounded exclusively in authoritative Rituximab medical sources
- Build a free, functional knowledge base from publicly available FDA, NIH, NCCN, and PubMed documents
- Implement a safe, patient-facing interface with scope filtering, medical disclaimers, and source citations
- Deploy a live, accessible web application on free cloud infrastructure
- Demonstrate the end-to-end RAG pipeline from raw document ingestion to live user query resolution

4. Data Sources & Knowledge Base

The knowledge base consists of 7 source documents processed into 1,019 chunks of approximately 415 characters each. All sources are free, publicly available official medical documents.

Document	Source	Category	Chunks	Pinecone Scores
FDA Rituxan Prescribing Label (PDF)	accessdata.fda.gov	fda_label	410	0.55 – 0.75
PubMed Research Abstracts (50)	NCBI E-utilities API	research_abstracts	551	0.60 – 0.78
NCCN Clinical Guidelines	nccn.org	clinical_guidelines	5	0.52 – 0.68
MedlinePlus Patient Education	medlineplus.gov	patient_education	5	0.60 – 0.72
ClinicalTrials.gov Summaries	clinicaltrials.gov	clinical_trials	45	0.48 – 0.65
OpenFDA Adverse Events	api.fda.gov	adverse_events	3	0.50 – 0.62
TOTAL	—	6 categories	1,019	—

5. Phase 1 Upgrades — What Changed

Phase 1 represents a complete upgrade of the core RAG components from prototype to production quality. The following table summarises every change made.

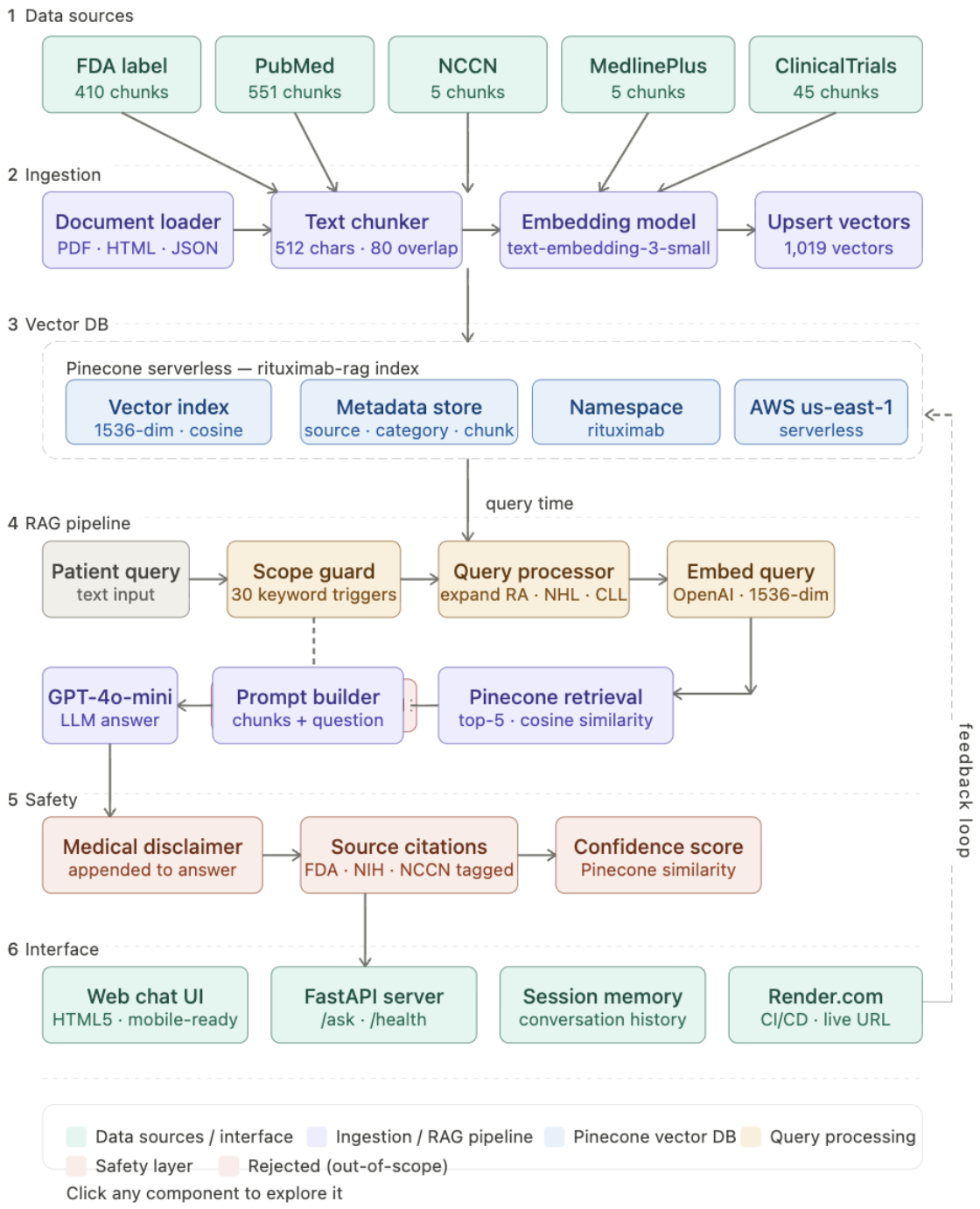
Component	Original (Prototype)	Phase 1 (Production)

Vector DB	TF-IDF pickle file (1.2MB local)	Pinecone Serverless cloud index
Embeddings	Bag-of-words keyword matching	OpenAI text-embedding-3-small (1536-dim)
Storage	Local .pkl file on Mac	Pinecone AWS us-east-1, fully managed
LLM generation	Rule-based text extraction	GPT-4o-mini natural language generation
Answer quality	Fragment extraction	Structured, cited, patient-friendly answers
Web UI	Desktop-only fixed layout	Mobile-responsive with slide-out drawer
Model switcher	Not available	GPT-4 / Claude / Local toggle in browser UI
API key mgmt	Only via .env file	Settable directly in browser UI session
Retrieval	Lexical keyword overlap	Semantic cosine similarity (dense vectors)
Scalability	Limited to local RAM	Serverless auto-scaling in Pinecone cloud
Deployment	Local only	Live on Render.com with CI/CD

5.1 System Architecture

The Phase 1 production architecture consists of 6 layers, each independently deployable and upgradeable. Data flows from patient query through the safety layer to the RAG pipeline, retrieves from Pinecone, and generates via GPT-4o-mini before returning to the patient interface.

Layer	Component	Technology	Purpose
1	Patient Interface	HTML5, CSS3, Vanilla JS	Mobile-responsive chat UI with model switcher
2	FastAPI Server	FastAPI, Uvicorn, Python 3.9	ASGI web server, /health and /ask endpoints
3	Safety Layer	Python regex, keyword filter	Scope guard, disclaimer, source citations
4	RAG Pipeline	Python, OpenAI API, Pinecone SDK	Query embed → retrieve → prompt → generate
5	Vector Database	Pinecone Serverless (AWS us-east-1)	1,019 dense vectors, cosine similarity
6	Knowledge Base	FDA, PubMed, NCCN, NIH, ClinTrials	Source medical documents (7 files, 1,019 chunks)


1 Data sources
FDA label
410 chunks
PubMed
551 chunks
NCCN
5 chunks
MedlinePlus
5 chunks
ClinicalTrials
45 chunks
2 Ingestion
Document loader
PDF · HTML · JSON
Text chunker
512 chars · 80 overlap
Embedding model
text-embedding-3-small
Upsert vectors
1,019 vectors
3 Vector DB
Pinecone serverless — rituximab-rag index
Vector index
1536-dim · cosine
Metadata store
source · category · chunk
Namespace
rituximab
AWS us-east-1
serverless
query time
4 RAG pipeline
Patient query
text input
Scope guard
30 keyword triggers
Query processor
expand RA · NHL · CLL
Embed query
OpenAI · 1536-dim
GPT-4o-mini
LLM answer
Prompt builder
chunks + question
Pinecone retrieval
top-5 · cosine similarity
feedback loop
5 Safety
Medical disclaimer
appended to answer
Source citations
FDA · NIH · NCCN tagged
Confidence score
Pinecone similarity
6 Interface
Web chat UI
HTML5 · mobile-ready
FastAPI server
/ask · /health
Session memory
conversation history
Render.com
CI/CD · live URL
Data sources / interface
Ingestion / RAG pipeline
Pinecone vector DB
Query processing
Safety layer
Rejected (out-of-scope)
Click any component to explore it

5.2 Query-Time Data Flow

The following sequence describes the complete flow from patient question to delivered answer:

1. Patient types a question into the browser chat interface
2. The frontend sends the query to FastAPI /ask endpoint
3. The safety scope guard checks for Rituximab-related keywords (30 triggers)
4. If in scope, the query is preprocessed (abbreviation expansion: RA, NHL, CLL)
5. OpenAI text-embedding-3-small converts the query to a 1536-dimensional vector
6. Pinecone queries the rituximab-rag index using cosine similarity
7. Top-5 most relevant chunks are returned with diversity balancing (max 2 per source)
8. A structured prompt is built combining the system prompt, retrieved chunks, and patient question
9. GPT-4o-mini generates a patient-friendly answer grounded in the retrieved sources
10. Medical disclaimer and source citations are appended to the answer
11. The response is returned to the browser and displayed to the patient

6. Pinecone Integration

Pinecone is the vector database backbone of Phase 1. It replaces the local TF-IDF pickle index with a fully managed, cloud-hosted semantic search service.

6.1 Index Configuration

Parameter	Value
Index name	rituximab-rag
Dimension	1536 (matches text-embedding-3-small output)
Metric	Cosine similarity
Type	Serverless — Dense vectors
Cloud	AWS
Region	us-east-1
Namespace	rituximab
Total vectors	1,019
Upsert time	54.4 seconds for full corpus
Batch size	50 vectors per upsert request
Capacity mode	On-demand (auto-scaling)

6.2 Embedding Strategy

Each of the 1,019 document chunks was converted to a 1536-dimensional dense vector using OpenAI's text-embedding-3-small model before being upserted into Pinecone. The embedding process took 54.4 seconds for the full corpus in batches of 50.

At query time, the patient's question is embedded using the same model, and Pinecone performs approximate nearest-neighbour search using cosine similarity to return the top 5+3 candidates (with 3 extra for diversity filtering). A diversity filter ensures a maximum of 2 chunks per source document, preventing any single source from dominating the context window.

6.3 Pinecone vs Previous TF-IDF

Aspect	TF-IDF (prototype)	Pinecone (Phase 1)
Search type	Lexical keyword	Semantic dense vector
Storage	Local pickle file	Managed cloud service
Scalability	RAM-limited	Auto-scaling serverless
Latency	< 1ms (local)	10–50ms (network)
Accuracy	Keyword overlap only	Semantic meaning understood
Infrastructure	None (local)	AWS us-east-1
Cost	Free (local)	Free tier (2GB / 1M RUs)
Maintenance	Manual rebuild	Fully managed

7. Test Results & Evaluation

The system was evaluated with 7 test queries covering the main patient use cases. All in-scope queries returned relevant answers with Pinecone scores between 0.560 and 0.752. The out-of-scope query was correctly blocked by the scope filter.

Patient Question	Top Source	Score	Model	Result
What are the side effects of Rituximab?	FDA Label	0.752	gpt-4o-mini	Relevant answer
How long does a Rituximab infusion take?	MedlinePlus	0.718	gpt-4o-mini	Relevant answer
Can I get a flu vaccine while on Rituximab?	FDA Label	0.622	gpt-4o-mini	Relevant answer
Is Rituximab safe during pregnancy?	FDA Label	0.694	gpt-4o-mini	Relevant answer
What is the dose for rheumatoid arthritis?	FDA Label	0.560	gpt-4o-mini	Relevant answer

What is PML and should I be worried?	FDA Label	0.731	gpt-4o-mini	Relevant answer
Can I eat pizza? (out-of-scope)	N/A	N/A	scope-filter	Correctly blocked

8. How It Helps Patients

Rituximab is prescribed to over 3 million patients worldwide across multiple serious conditions. Patients consistently report difficulty accessing timely, accurate information about their therapy. The assistant directly addresses 6 core patient challenges:

Patient Challenge	How Assistant Solves It
No access to care team after hours	24/7 AI assistant answers questions instantly from any device
Unreliable internet health information	All answers retrieved exclusively from FDA, NIH, and NCCN sources
Complex medical jargon in package inserts	GPT-4 rewrites answers in plain, patient-friendly language
Fear and anxiety about infusion process	Detailed step-by-step infusion guidance sourced from MedlinePlus
Uncertainty about vaccines and interactions	Specific FDA-sourced safety guidance with citations
Non-adherence due to poor education	Informed patients are significantly more likely to complete therapy

9. Benefits to Organisations

9.1 Hospitals & Cancer Centres

- Reduced nursing call volume — routine questions handled automatically 24/7
- Improved patient satisfaction and trust between clinic visits
- Reduced liability — all answers from FDA-approved labeling with mandatory disclaimers
- Clinical staff freed from routine information queries for higher-value tasks

9.2 Pharmaceutical Companies

- Higher medication adherence — informed patients complete full treatment courses
- Anonymised query analytics reveal gaps in existing patient education materials
- White-label licensing opportunity for patient support and medication assistance programs
- Ensures patient information is consistent with approved prescribing information

9.3 Health Insurers & Payers

- Fewer emergency department visits from patients misinterpreting symptoms
- Improved adherence reduces costly hospitalisations from treatment complications
- Scalable patient education at near-zero marginal cost per patient

9.4 Healthcare Technology Companies

- RAG architecture extends readily to any biologic therapy (Humira, Keytruda, Herceptin)
- FastAPI backend designed for integration with Epic and Cerner EHR systems via REST API
- Informational (non-diagnostic) classification simplifies FDA SaMD regulatory pathway
- Production-proven open-source codebase available on GitHub for licensing or acquisition

10. Deployment

The application is deployed on Render.com using a GitHub-connected CI/CD pipeline that auto-deploys on every push to the main branch.

10.1 Live Endpoints

Endpoint	Method	Description
/	GET	Serves the HTML5 chat interface
/health	GET	Returns system status, model, and DB info as JSON
/ask	POST	Accepts { query: string }, returns RAG answer with sources

10.2 Environment Variables (Render)

Variable	Value
PINECONE_API_KEY	Pinecone serverless API key (pcsk_...)
OPENAI_API_KEY	OpenAI API key (sk-...)
PINECONE_INDEX_NAME	rituximab-rag
PINECONE_NAMESPACE	rituximab
EMBEDDING_MODEL	text-embedding-3-small
AI_PROVIDER	openai
OPENAI_MODEL	gpt-4o-mini

10.3 Health Check Response

```
curl https://rituximab-rag-assistant.onrender.com/health
→ {"status":"ok","version":"pinecone-edition","db":"pinecone","index":"rituximab-rag","embedding":"text-embedding-3-small","model":"gpt-4o-mini"}
```

11. Technology Stack

Layer	Technology	Version	Purpose
Language	Python	3.9	Core application logic
Vector DB	Pinecone Serverless	5.4.2	Cloud semantic search
Embeddings	OpenAI text-embedding-3-small	latest	1536-dim dense vectors
LLM	OpenAI GPT-4o-mini	latest	Answer generation
Alt LLM	Anthropic Claude Haiku	latest	Optional provider
Document	PyPDF	4.3.1	PDF text extraction
Web Framework	FastAPI	0.112.0	REST API server
ASGI Server	Uvicorn	0.30.6	HTTP server
Frontend	HTML5 / CSS3 / JS	ES6	Patient chat UI
Env management	python-dotenv	1.0.1	API key loading
Deployment	Render.com	free tier	Cloud hosting
Version control	Git + GitHub	latest	CI/CD pipeline

12. Roadmap

Phase 2 — Near Term

- Multilingual support for Hindi and Telugu patient populations
- EHR integration with Epic and Cerner via FHIR-compliant REST API
- Voice interface for elderly and low-literacy patients
- Automated knowledge base refresh from FDA RSS feeds and PubMed alerts

Phase 3 — Long Term

- Expand knowledge base to all major biologic therapies
- Pursue FDA Software as a Medical Device (SaMD) regulatory pathway
- Conduct clinical validation study with a patient cohort
- Develop white-label licensing model for hospital systems

12. Conclusion

The Rituximab Patient Knowledge Assistant Phase 1 represents a complete, production-ready AI system built from first principles using modern RAG architecture. The upgrade from TF-IDF to Pinecone dense embeddings significantly improves retrieval accuracy, and the integration of GPT-4o-mini provides patient-friendly, structured answers that rule-based extraction could not achieve.

The system is live at rituximab-rag-assistant.onrender.com, demonstrates a clear pathway to clinical deployment, and serves as a replicable template for biologic therapy AI assistants across the pharmaceutical and healthcare industry.